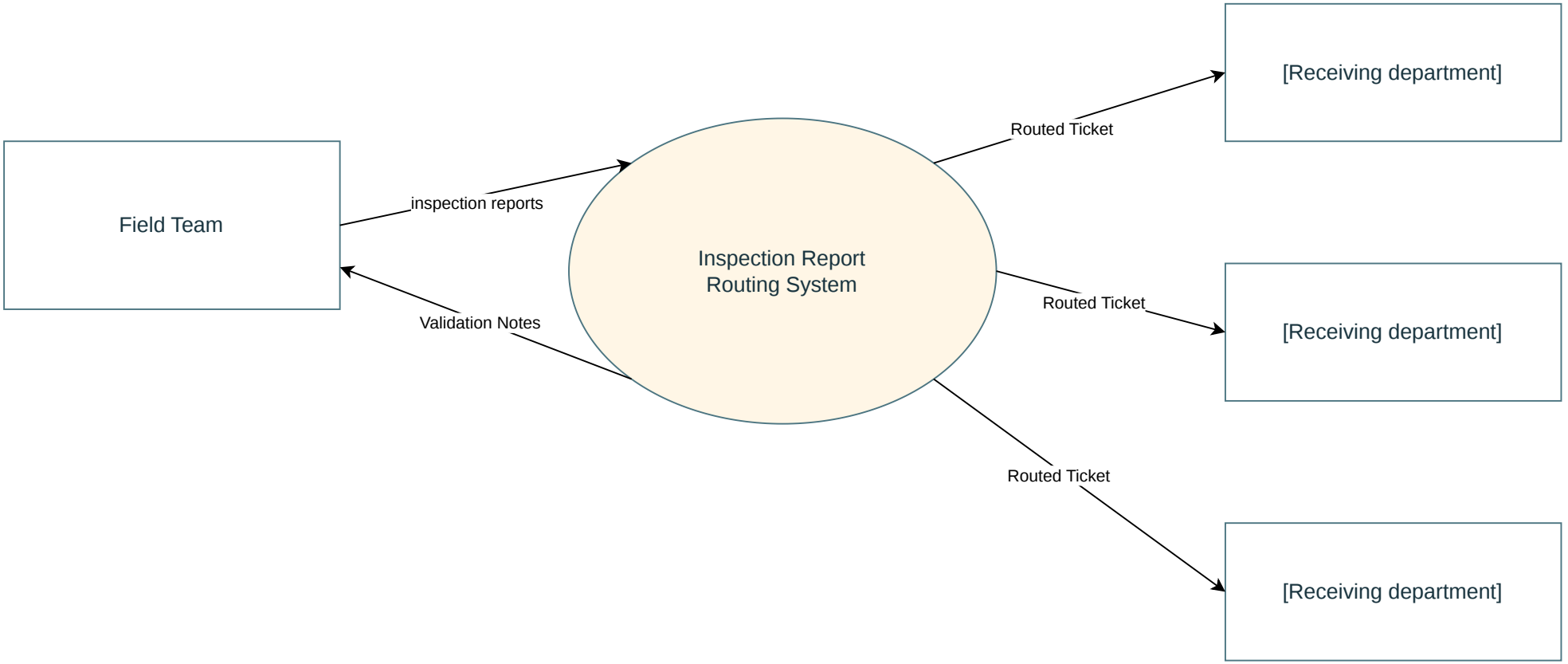
CIT 140 · Week 6 · System boundary

Replace prompts. Use W06-S01 for the system, external teams and data crossing the boundary. Keep internal stores on page 2. Add labeled connectors.



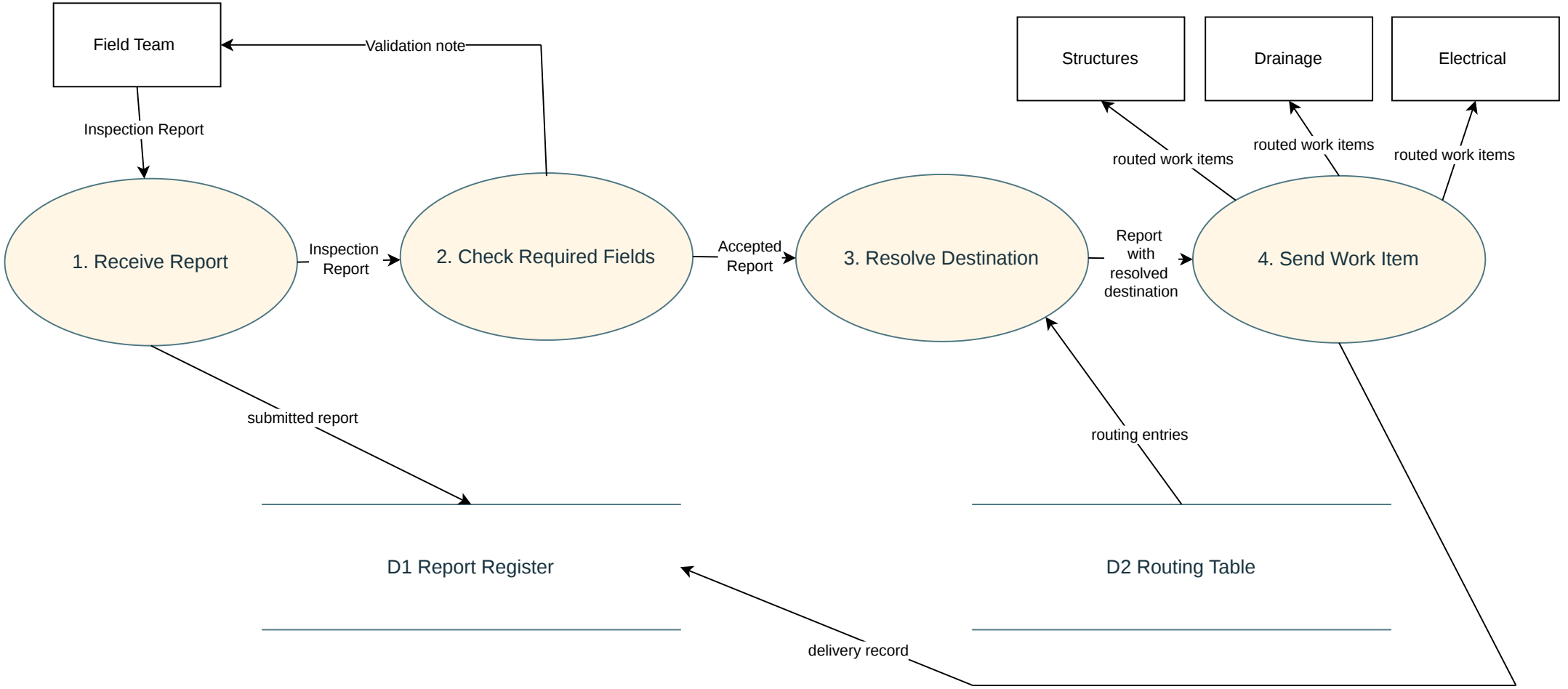
Inside: field team receiving inspection reports, validation notes from inspection report routing system. electrical, drainage, and structures receiving routed work items from inspection routing system

Outside: Flight control, physical inspections, and repair work

Sources: W06-S01

CIT 140 · Week 6 · Information movement

Use W06-S01. Connect the four processes, two stores and external teams. Label arrows with data; balance external flows with page 1. The starter positions are a layout aid, not a sequence diagram.



Receive Report takes inspection report from field team, then sends submitted report to d1 report register, and a inspection report to check required fields. Required fields check if all 8 required fields are there, sends validation note if they are and sends accept report to resolve destination. Resolve Destination reads routing from routing table and passes report with destination to send work item. Send work item sends the work item to the correct department and writes a delivery record to D1 report register.

CIT 140 · Week 6 · Definitions and finding

Complete every column for the eight fields. Replace prompts with short answers. Use the same fields as W06-S02.

Field	Meaning	Example	Allowed value / format	Source
report_id	unique report reference	CJ-603	CJ- folllowed by three digits	W06-S02
asset_id	reference for inspected asset	BR-301	two letter prefix and three digits	W06-S02
asset_type	type of report	BRIDGE	Either STREETLIGHT, DRAIN, OR BRIDGE	W06-S02, W06-S03
finding_code	code for type of issue	SURFACE_DAMAGE	one of LAMP_OUT, BLOCKED, or SURFACE_DAMAGE	W06-S02
priority	level of urgency	ROUTINE	ROUTINE or URGENT	W06-S02
captured_at	Capture time	2026-10-14T09:10:00Z	YYYY-MM-DDTHH:MM:SSZ	W06-S02
photo_ref	attachment reference	IMG-603	IMG- followed by 3 digits	W06-S02
survey_area	fictional areas	CENTRAL	one of NORTH, CENTRAL, or, SOUTH	W06-S02

Finding: [Report ID, expected department, observed department, source IDs. Use 2–3 sentences.] In W06-S02, CJ-603, CJ-604, AND CJ-607 all have asset types that are of DRAIN and their required destination is the drainage department according to the approved destination policy in W06-S03 but they were not sent to that department. In the delivery logs in W06-S04, it shows that these reports were received by electrical every time which shows issues in incorrect routing.

Unanswered question: [What evidence would help establish the cause?] Something that would help understand the cause would be to check how resolve destination is reading routing entries from D2 Routing Table and if the D2 routing table is giving correct information to the resolve destination. This would need to be checked with the service administrator that maintains the table.